



Sevan Evans

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EDUCATION

Associates of Science, Physics

College Of The Desert, Palm Desert, CA

Expected May 2027

Activities & Clubs:

- Founder & President, Physics Club (www.codphysicsclub.com), 40+ members, secured \$3,000 in grants for rocketry)
- NASA NCAS Robotics Competition (Awarded Team MVP)
- MESA (First generation STEM students) Program & Club Member

SKILLS

Scientific Computing: Three.js, Python, MATLAB, Mathworks Simulink

CAD: Fusion 360, OnShape

Front-End & UI Development: React, TypeScript, HTML/CSS, Kotlin Compose

System Integration: REST, WebSocket, Firebase, WebRTC

Development Tools: Git, CI/CD, Linux

Backend & Infrastructure: Node.js, Express.js, Google Cloud, AWS, MongoDB

Additional Languages: Java, C++, C#, Swift, Lua, Bash scripting

WORK EXPERIENCE

Autonomous Rover Hazard Detection Intern | *NASA* | NASA Ames, CA | Full-Time

Aug 2026 - Present

- Optimize YOLO-based hazard detection algorithms for real-time performance on computationally-constrained lunar rovers through model compression, quantization, and architecture tuning; evaluated performance under varied lunar lighting and dust conditions to advance autonomous surface safety.

Student Researcher | *NASA M-STAR* | **College of the Desert, CA**

Mar 2026 - Present

- Conducted 30 experiment cycles in a NASA-funded laboratory extracting oxygen and hydrogen from naturally found Martian regolith and salt water via electrolysis, logging procedural data and analyzing gas chromatography readings to quantify yield efficiency for life-support systems and propellant production supporting Mars sample return and future human habitats.

STEM Teacher | *WindTree Education* | Palm Desert, CA | Part-time

Sept 2025 - May 2026

- Instruct classes of 30+ K–12 students in foundational engineering concepts using hands-on demonstrations of robotics, mechatronics, chemistry, astronomy, and physics experiments.

Front-End Software Engineer | *Quick Draw Development* | Remote | Contract

Jan 2024 - Feb 2024

- Built an advanced multi-criteria data-filtering system for a large-scale industrial processing platform, enabling high-volume dataset analysis and traceability for over 100 enterprise clients.

End User Software Engineer | *Tigase, Inc* | Remote | Full-time

Jan 2021 - Dec 2024

- Architected and developed the front-end of Tygrys, a custom XMPP communications application for privately hosted messaging servers using Kotlin Compose Multi-platform with deployments on Android, iOS, Linux, Windows, and macOS.
- Set up and maintained automated CI pipelines using OneDev, enabling automated testing and deployment; optimized builds and cut artifact creation time by over 95%.

Full-Stack Software Engineer | *BodyMX* | Palm Springs, CA | Full-time

Sept 2022 - Apr 2024

- Co-founded and engineered a real-time React, Typescript, personal wellness platform integrating WebRTC video, telemetry logging, full user-role hierarchy, and hybrid backend with Firebase + MongoDB, driving 50% MoM app traffic growth

Repair Technician | *WeFixx* | Palm Springs, CA | Part-time

Aug 2020 - Jan 2021

- Repaired over 50 smartphones and laptops with 100% customer satisfaction rate, handling a wide variety of issues including screen replacements, battery swaps, water damage repairs, and software troubleshooting.

FEATURED TECHNICAL PROJECTS

JumperSim | <https://github.com/plaidroni/JumperSim> - Live Demo: www.jumpersim.com

2025

Physics-based 3D Skydiving Simulator & Safety Training Tool

A 3D skydiving simulation using Three.js and Typescript that models realistic jumper dynamics in freefall and under canopy, with real-time weather, wind layering, and full, scrubbable playback controls. Designed to predict and prevent canopy collisions

- Created a custom physics engine to simulate skydiver trajectories affected under layered wind forces at 1,000 ft AGL intervals using fetched real-time Open-Meteo data.
- Built-in visualization tools utilized directly by the United States Parachute Association for a visual separation awareness minigame, supporting canopy collision prevention training
- Developed user-friendly GUI with draggable panels, full timeline scrubbing capabilities, and dynamic Mapbox satellite integration, supporting multiple aircraft types with accurate flight characteristics and jump run simulations

Model Rocket Thrust Vectoring Control System | <https://github.com/plaidroni/Model-Rocket-Flight-Computer>

2025

- Engineered a 2-axis Thrust Vector Control system on an Arduino Nano (C++), integrating IMU sensor fusion (accelerometer, gyroscope, magnetometer, barometer) with a finely tuned PID controller driving a closed-loop attitude correction, on a mount designed in Fusion 360, with the long-term goal of achieving autonomous propulsive landing.

SPECIALIZED EXPERIENCE

- **NASA Community College Aerospace Scholars Robotics Competition**
 - Selected participant in an intensive 5-day robotics competition designing and building rovers under strict time and resource constraints
 - Awarded team MVP by mentor for collaborative leadership, soft skills, and technical ingenuity
- **Active B-licensed skydiver and member of the United States Parachute Association**
 - Demonstrates high-stakes decision-making, risk assessment, and clear communication under pressure.
 - Actively Pursuing FAA Riggers Certification